UNIT V – Chemical, Electrochemical & Thermal Energy Based Processes

23A03501 MACHINING PROCESSES · B.Tech Mechanical · JNTUK

1. Chemical Machining (CHM)

CHM = controlled **chemical dissolution** (etching) of metal from selected areas using a strong **etchant**; other areas protected by **maskant**.

- No tool, no force, no heat → stress-free, burr-free surface
- Material removed by **corrosive attack** (reaction: metal + etchant → soluble salt + H_2)
- Types:
  - **Chemical milling** – shallow cavities, weight reduction (large panels)
  - **Chemical blanking** – cut thin sheet (<0.5 mm) through, intricate shapes
  - **Photochemical machining (PCM)** – maskant applied by photo-resist; used for PCBs, fine meshes
  - Also: chemical engraving, chemical deburring

Steps in chemical milling

1. **Cleaning** – remove oil, grease, scale (solvent / alkali) → good maskant adhesion
2. **Masking** – coat whole part with maskant (dip / spray / brush), cure
3. **Scribing** – cut & peel maskant from areas to be machined (template)
4. **Etching** – dip in etchant tank (temp-controlled, agitated)
5. **Rinsing & demasking** – wash, strip maskant, final clean

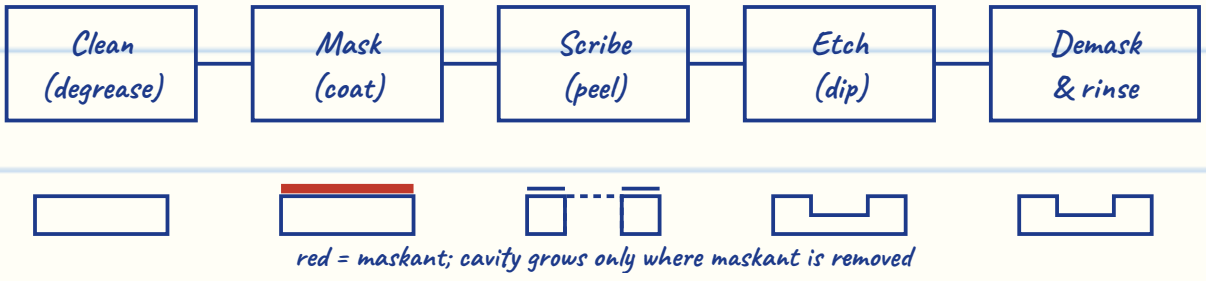


Fig 1. Sequence of chemical milling

Maskants (resists)

- Must: stick well, resist etchant, easy to scribe/strip, no reaction with metal

Maskant	Use / remark
Neoprene (rubber)	Dip/spray; most common for chem milling, 0.2–0.5 mm film; Al, Mg, Ti
PVC (vinyl)	Strippable; good for acid etchants
Polyethylene	Resist to strong acids; sheet/film form
Photoresist	Light-sensitive (KPR); fine details (±0.01 mm) → PCM

Techniques of applying maskants

- **Cut-and-peel** – dip/spray coat → cure → template placed → cut with knife → peel region by hand. Large parts, low volume, accuracy ±0.1 mm
- **Screen printing** – maskant pushed via silk/steel screen stencil; pattern printed directly; no scribing; medium accuracy, high volume (small flat parts)
- **Photoresist (photographic)** – coat with photoresist → expose UV through **negative** → develop → bake; polymerised area stays. High accuracy, fine/thin parts

Photochemical machining (PCM) steps

- Artwork (master) → photo-tool negative → coat sheet with resist → UV expose → develop → etch (spray $FeCl_3$) → strip resist

Etchants

Work material	Etchant	Remark
Aluminium	NaOH (10–20%) / $FeCl_3$	NaOH ~ 80 °C
Copper	$FeCl_3$ (also ammonium persulphate)	PCB making
Steel	HNO_3 / HCl (+ $FeCl_3$)	Fe-chloride for stainless
Titanium	HF + HNO_3 (HF 10–20%)	H_2 pick-up risk
Magnesium	H_2SO_4 / HNO_3 dilute	Fast attack

- Mnemonic: **Al–NaOH, Cu–$FeCl_3$, Ti–HF, Steel–acids**

Etch rate, undercut, etch factor

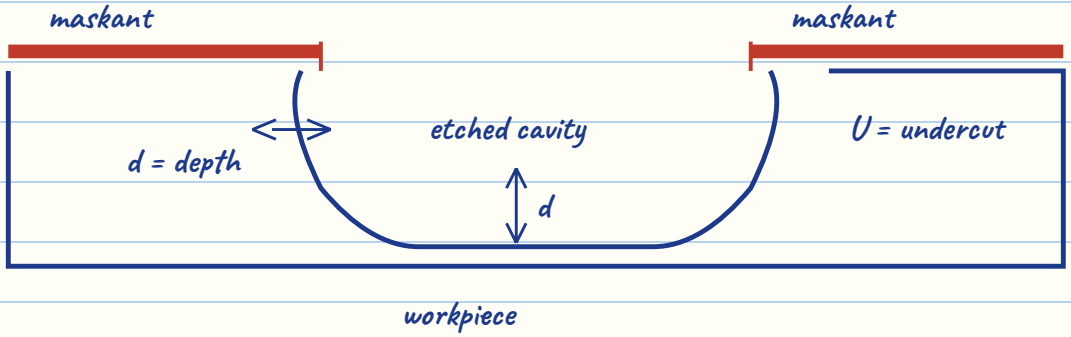


Fig 2. Undercut under the maskant (etching is isotropic)

- Etching is **isotropic** → sideways attack = **undercut (U)**

> Etch factor Fe = U / d (or d / U)

> Etch rate = depth / time (mm/min) ; U ≈ d (for Fe≈1)

- Typical etch rate 0.02–0.04 mm/min; depth 0.01–12 mm; tolerance ±0.05–0.1 mm
- Time t = d / etch rate; MRR = (area A × depth d) / t = A × etch rate
- Small etch factor (better) ⇒ sharper walls; **corner radius ≈ depth**

Process parameters

- Etchant concentration & temperature (±1–2 °C), agitation, maskant thickness, material composition/grain, surface finish (0.5–2.5 µm Ra)

Applications, advantages, limitations

Applications	Advantages	Limitations
Aircraft skin panels, wing/fuselage weight reduction; PCBs; nameplates; turbine blades; thin foils, meshes; deburring; micro-parts	No tool wear / force; no burrs, no HAZ; complex shapes; many parts at once; low capital; any hard material	Slow (min/mm); only shallow depth; undercut → low accuracy; handling of corrosive/toxic chemicals, disposal; porosity / grain attack; maskant cost; hydrogen embrittlement

2. Electrochemical Machining (ECM)

ECM = reverse of electroplating. Workpiece (**anode +**) dissolves anodically in a flowing **electrolyte** when DC passes across a small gap to the shaped **tool (cathode −)**.

- Based on **Faraday's laws of electrolysis**
- No contact, no tool wear, no heat damage; tool shape copied into work (negative copy)
- Electrolyte carries away metal hydroxide sludge & heat & H_2 gas
- Reactions (Fe in NaCl): $Fe \rightarrow Fe^{2+} + 2e^-$; $2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$; $Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2\downarrow$ (sludge)

Faraday's laws

- **I law:** mass dissolved m ∝ charge Q = I·t
- **II law:** m ∝ equivalent weight E = A/Z (A = atomic wt, Z = valency)

> $m = (A \cdot I \cdot t) / (Z \cdot F)$ [F = 96500 C/mol]

Derivation of MRR

1. With current efficiency η: $m = \eta A I t / (Z F)$
2. Mass rate $\dot{m} = m/t = \eta A I / (Z F)$ (g/s)
3. Divide by density ρ → volume rate:

$$MRR = \eta A I / (\rho F Z) \ cm^3/s$$

- Also MRR = (A/ZF)(η)(J·a) with **J = current density**, a = area; and MRR = a × f (f = feed rate) ⇒ **$f = \eta AJ/(\rho F Z)$**
- Alloy: use $1/E = \Sigma(x_i \cdot Z_i/A_i)$ form: equivalent weight of alloy $E = 1/\Sigma(x_i Z_i/A_i)$, x_i = wt fraction

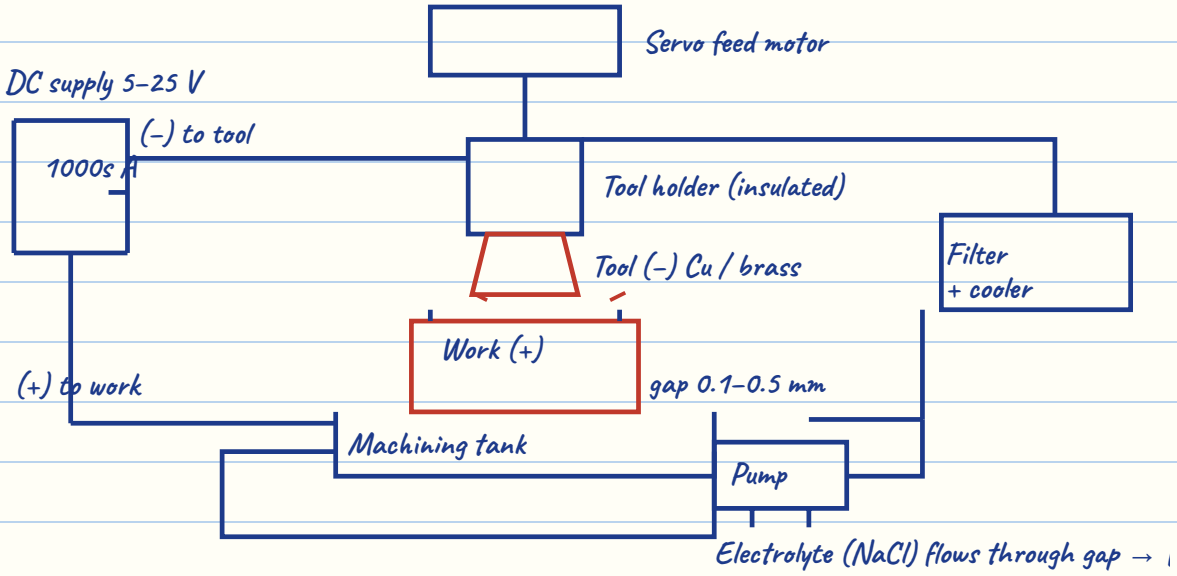


Fig 3. Schematic of ECM set-up

Equipment (main units)

- **Power supply** – low-V (5–25 V), very high-I (50–40000 A) DC, ripple-free
- **Tool feed system** – servo motor/hydraulic, constant feed 0.1–20 mm/min, short-circuit protection
- **Electrolyte system** – tank, pump (flow 10–60 m/s in gap, 1–3 MPa), filter/centrifuge, heat exchanger, pH control
- **Work-holding fixture** (insulated), machine tank, tool holder

Tool materials & tool design

- Tool material: **Copper, brass, stainless steel**, Ti, Cu-W (good conductivity, machinable, corrosion resistant, stiff)
- Tool is **never worn** (only cathode, H_2 evolves)
- Tool design aspects:
  - Cathode shape ≠ exact work shape → **overcut** compensation (tool made smaller by gap)
  - Insulate tool sides (epoxy / Teflon) → avoid stray machining

- Provide flow holes / passages for uniform electrolyte flow; avoid dead zones & sharp corners
- Rigid, avoids vibration; tool shape found by trial-and-error / CAD (cut-and-try)

Electrolytes

➤ Needs: high conductivity, non-toxic, stable, non-corrosive to machine, low cost, controls passivation

Electrolyte	Features
NaCl (10–30%)	Cheap, high conductivity, η ~100%, Fe/steel; poor accuracy (stray cut), corrosive
$NaNO_3$ (10–25%)	Passivating → better accuracy (less stray); η depends on J; costlier; used for precision parts
NaOH / KCl / $NaClO_3$	Special uses ($NaClO_3$ - precision dies)

Process parameters

Parameter	Typical value / effect
Voltage	5–25 V DC
Current density J	10–100 A/cm^2 (up to 500); MRR $\propto J$
Inter-electrode gap	0.1–0.5 mm; smaller gap ⇒ better accuracy, higher J
Feed rate	0.1–20 mm/min (typ 1–2)
Electrolyte flow/temp	~ 20–60 °C, 5–50 m/s, pressure 0.5–3 MPa
MRR	≈ 1.6 mm^3/s per 100 A (steel); surface finish 0.2–1.5 µm; accuracy ±0.02–0.1 mm

Equilibrium gap

➤ At start gap large → J low → MRR low; as gap shrinks J rises → rate rises. Gap stabilises where **dissolution speed = tool feed f** → **equilibrium gap h_e**

➤ $J = \kappa V/h$ (κ = electrolyte conductivity, V = effective voltage after drops)

$$f = \eta A \kappa V / (\rho F Z h) \Rightarrow h_e = \eta A \kappa V / (\rho F Z f)$$

➤ $h_e \propto V, \kappa$; $\propto 1/f$ → **self-regulating** process (increase f ⇒ gap decreases)

★ Important

➤ MRR = $\eta A I/(\rho F Z)$ – write derivation + units

➤ Tool wear ≈ zero; gap 0.1–0.5 mm; V 5–25 V; NaCl electrolyte

➤ ECM suits hard, tough, conductive metals – **not** insulators

Numerical (ECM)

Q. Iron ($A = 55.85$, $Z = 2$, $\rho = 7.86\ g/cm^3$) machined at $I = 1000\ A$, $\eta = 90\%$. Find MRR.

Sol. $MRR = \eta AI/(\rho FZ) = (0.9 \times 55.85 \times 1000)/(7.86 \times 96500 \times 2)$

$= 50265 / 1{,}516{,}980 =$ **$0.0331\ cm^3/s \approx 1.99\ cm^3/min$** ($\approx 33\ mm^3/s$)

Mass rate $= 0.9 \times 55.85 \times 1000/(2 \times 96500) = 0.26\ g/s$. If face area $= 2\ cm^2 \Rightarrow$ feed $f = 0.0331/2 = 0.0166\ cm/s \approx$ **10 mm/min.**

Applications	Advantages	Limitations
Turbine blades, jet-engine parts, die & mould sinking, deep holes, cavities, gun barrel rifling, deburring, contoured shapes in superalloys	High MRR for hard metals; no tool wear; no thermal damage / residual stress; burr-free; good finish; complex shapes in one pass; hardness irrelevant	Only conductive work; high power & capital cost; tool design costly; corrosion, sludge disposal; stray cutting, sharp corners hard; large floor space

3. Electro Chemical Grinding (ECG)

ECG = ECM + mechanical grinding. A **metal-bonded wheel** (conducting, with **insulating abrasive** – Al_2O_3 or diamond, CBN) is the cathode (–); work is anode (+). Electrolyte ($NaNO_3$ / NaCl) flows into the gap.

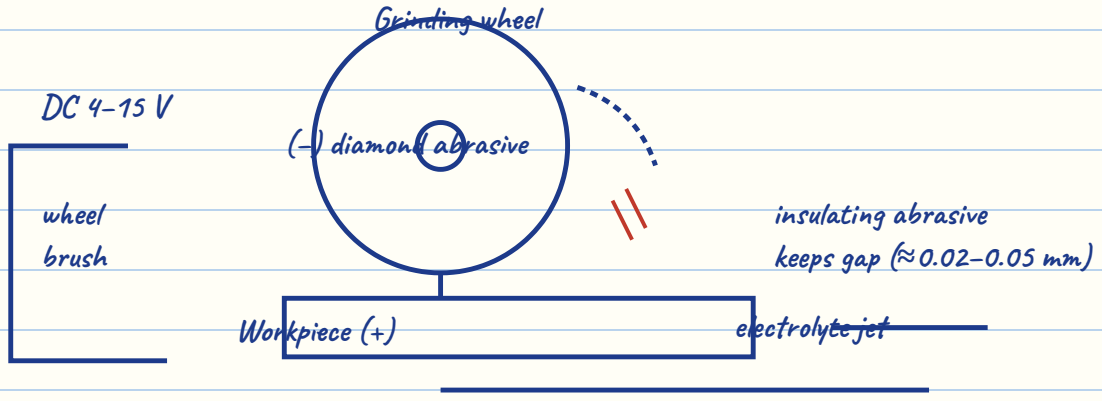


Fig 4. Electrochemical grinding

- Abrasive grains act as **insulating spacers** (gap ≈ 0.025–0.05 mm) & remove the oxide film/passive layer
- MRR split: **~90% by electrolysis, ~10% by abrasion** → wheel wear very low (up to 10× less than conv. grinding)
- Voltage 4–15 V, $J \sim 120$–$240\ A/cm^2$; wheel speed ~ 30 m/s; electrolyte $NaNO_3$ 0.25 kg/l

- Used for carbide tools, hard alloys, thin-walled/ brittle parts, burr-free sharpening
- Adv: no burr, no heat cracks, long wheel life, good finish (0.2 μm); Lim: only conducting work, corrosion, costly set-up, slower for soft metals

4. Electro Chemical Honing (ECH)

- **ECH** = ECM + honing: **honing stones** (abrasive bonded) fixed on a **cathodic tool** rotating & reciprocating in a bore; electrolyte ($NaNO_3$) flows through
- Principle: anodic dissolution (~95%) removes material; stones (low pressure) **wipe off the passive oxide film** & correct geometry
- Tool = hone with insulated metal strips + abrasive sticks; bore size ±; cycle 1–2 min
- Applications: gear teeth, cylinder bores, hydraulic sleeves, hardened steel parts

Feature	ECG	ECH	Conventional honing
Tool	Rotating wheel	Hone with stones	Hone with stones
Surface	Flat, form, external	Internal bore/gear	Bore
Removal	90% electro-chem	~95% electro-chem	100% abrasion
Stone wear	Very low	Very low	High
Heat/burr	None	None	Some heat, stresses

CHM vs ECM

Item	CHM	ECM
Principle	Chemical dissolution	Anodic dissolution (electrolysis)
Energy	Chemical, no current	Electrical DC, high current
Tool	None (maskant)	Shaped cathode
MRR	Low (0.02 mm/min)	High (1–20 mm/min)
Accuracy	±0.05–0.1 mm, undercut	±0.02–0.1 mm
Depth	Shallow	Deep cavities
Cost	Low capital	High capital
Work	Any metal	Electrically conducting

5. Laser Beam Machining (LBM)

LASER = Light Amplification by Stimulated Emission of Radiation. LBM: highly focused, monochromatic, coherent beam melts & vaporises material (power density ~10^6–10^9 W/cm^2).

Laser principle

- Atom absorbs energy → excited state (**pumping**)
- **Population inversion**: more atoms in upper (metastable) level than lower
- **Spontaneous emission** → photon; photon hits excited atom → **stimulated emission** (identical photon: same λ, phase, direction)
- Mirrors (100% + partial) in a resonator → amplification → coherent beam out

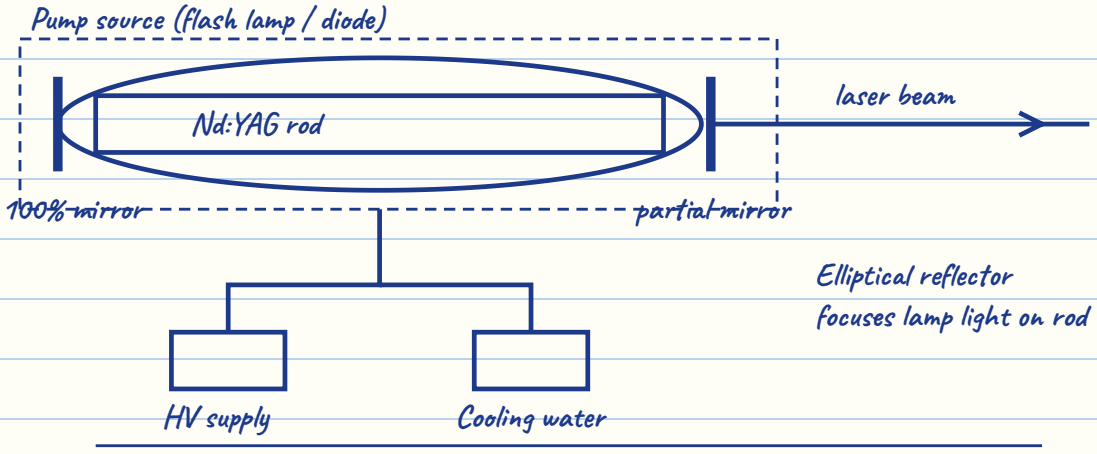


Fig 5. Nd:YAG solid-state laser

Types of lasers used

Laser	Medium / λ	Mode & power	Use
Ruby	$Al_2O_3 + Cr^{3+}$, 0.694 μm (red)	Pulsed; ~ 1 J, low eff. (<1%)	Old; spot welding, drilling
Nd:YAG	Nd^{3+} in YAG crystal, 1.064 μm (IR)	Pulsed / CW; 50 W–several kW; eff. 1–3%	Drilling, micro-welding, marking, thin cut
CO_2	$CO_2 + N_2 + He$ gas, 10.6 μm	CW/pulsed; 0.1–25 kW; eff. 10–15%	Cutting sheet, non-metals, welding

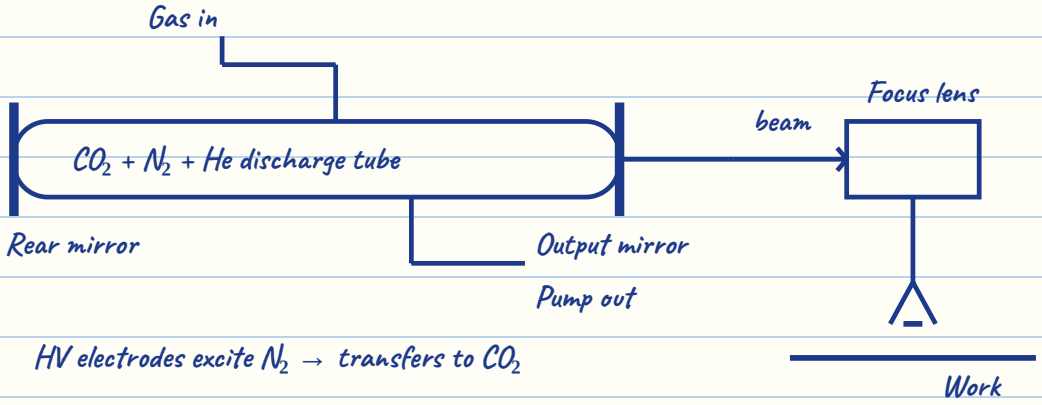


Fig 6. CO_2 laser & focusing on work (assist gas nozzle)

LBM set-up

- Laser source + power supply → beam delivery (mirrors / fibre) → focusing lens (+ assist-gas nozzle) → CNC work table. Shutter & beam monitor for control

Process parameters

- **Power / energy** (W; J per pulse): ↑ power ⇒ ↑ depth & kerf
- **Pulse duration** (ms–ns) & frequency: short pulse ⇒ less HAZ, vaporisation dominant
- **Beam spot diameter** $d = 2.44\,\lambda\, f / D$ (≈ 0.01–0.1 mm) → smaller spot, higher power density
- **Focal length** f & depth of focus: short f ⇒ small spot but shallow focus; position focus on/below surface
- **Assist gas** (O_2 – reactive, boosts cutting steel; N_2/Ar – inert, clean edge, stainless/Al; air) at 2–10 bar blows molten metal
- Work properties: reflectivity (Cu, Al reflect IR), thermal conductivity, absorptivity

$$\text{Power density} = P / (\pi\, d^2/4)\ (W/cm^2)\ ;\ \text{Energy per pulse } E = P \times t_p$$

Laser drilling & cutting

- **Drilling** – pulsed Nd:YAG; hole by melt + vaporisation + expulsion; Ø 0.01–1.0 mm, aspect ratio up to 20:1;
  - Methods: single-pulse, **percussion** (many pulses), **trepanning** (beam circles → larger holes), helical
- Applications: cooling holes in turbine blades, fuel-injector nozzles, wire-drawing dies, watch jewels, inkjet
- **Cutting** – CO_2 + assist gas; kerf 0.1–0.5 mm; steel up to 20–25 mm; speed up to m/min; CNC profiling, sheet metal, textiles, wood, plastics

- Other: welding, marking/engraving, surface hardening, micro-machining, scribing of wafers

Advantages	Limitations
Non-contact, no tool wear; any material (even diamond, ceramics); tiny precise holes; easy automation; small HAZ for short pulse	High capital & low efficiency; taper in holes; HAZ, recast layer, microcracks; depth limit (~ < 10–20 mm); reflective materials difficult; safety (eye hazard)

6. Plasma Arc Machining (PAM)

Plasma = 4th state of matter: ionised gas (ions + electrons). **PAM** – constricted high-velocity plasma jet at **10,000–30,000 K** (gas speed ~ supersonic) melts & blows away metal.

- Gas (Ar, H_2, N_2, air or mixtures) heated by arc between tungsten cathode & nozzle/work → ionises → plasma
- Gas forced through a **constricting water-cooled copper nozzle** → arc narrow, density & temp ↑
- **Gases:** Ar–H_2 (35% H_2) for thick Al/SS; N_2 (cheap) for general; air for carbon steel; Ar for inert

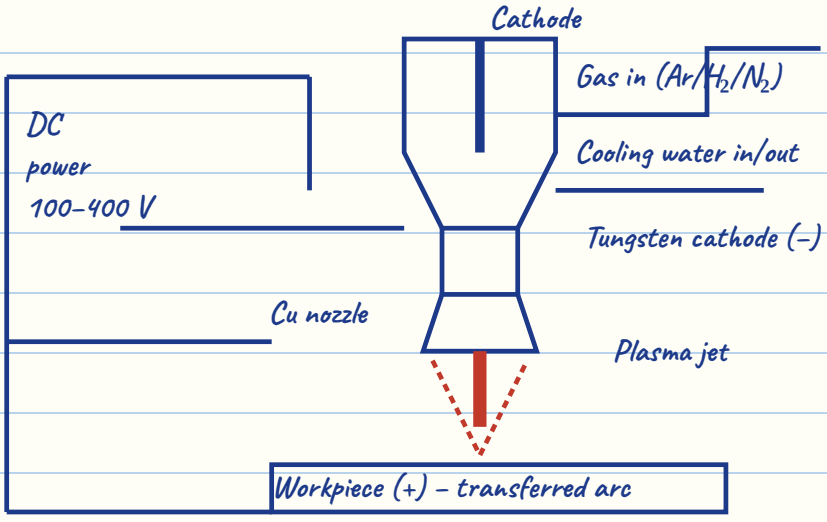


Fig 7. Plasma torch – transferred arc type

Transferred vs non-transferred arc

Item	Transferred arc	Non-transferred arc
Arc between	Electrode and **workpiece**	Electrode and **nozzle** (inside torch)
Work	Part of circuit (conductive)	Not in circuit; any material
Heat to work	High (anode heating, ~ 70% efficient)	Only plasma-jet heat (lower)
Use	Cutting, welding, metals	Spraying, non-conductors, heating

Process parameters

- Arc current 50–1000 A (typ 200–600 A), voltage 30–250 V, power 2–200 kW
- Gas type & flow (≈ 2–10 m^3/h), nozzle dia 1–5 mm, stand-off 5–10 mm
- Cutting speed (0.2–5 m/min) ↓ with thickness; torch angle; kerf 2–5 mm

Applications / HAZ

- Cutting **stainless steel, aluminium**, Cu, Ni alloys, cast iron (up to 150 mm); profile cutting of plate; shipbuilding; hard-facing; plasma spraying; turning of hard alloys (plasma-assisted)
- **HAZ**: ~ 0.5–1.5 mm — heat-affected layer + recast; taper & rough edge; may need finishing

Advantages	Limitations
Very fast cutting (≈ 10× oxy-fuel on SS/Al); cuts any conducting metal; low cost; good for thick plate; automation easy	Large kerf & HAZ; taper; metallurgical damage (cracks, oxide/nitride layer); noise, UV, fumes; poor accuracy (±1 mm); high power consumption

7. Electron Beam Machining (EBM)

EBM – a focused stream of **high-velocity electrons** (≈ 0.5–0.8 c) strikes work in **vacuum**; K.E. → heat → local melting & vaporisation. Pulsed beam drills micro-holes.

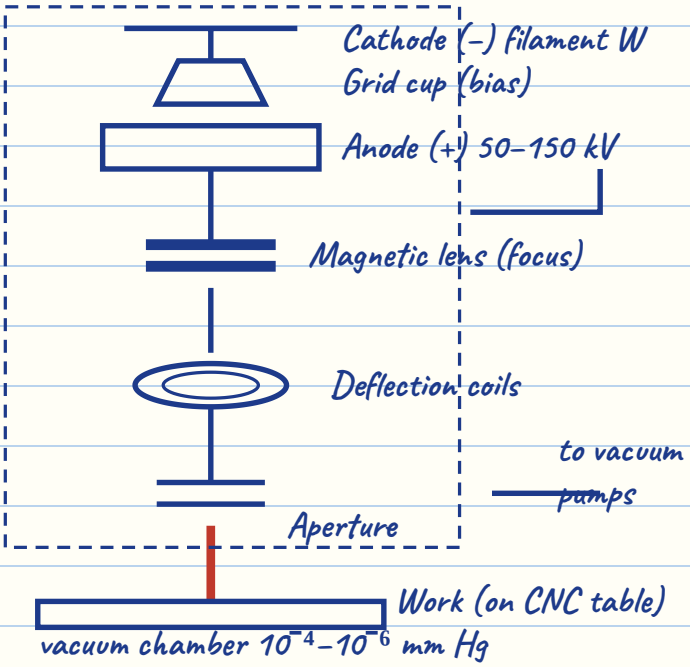


Fig 8. Electron-beam gun & column

Working

1. Tungsten filament heated → **thermionic emission**
2. Electrons accelerated by anode (**50–150 kV**) to ~ 1/2 speed of light
3. **Magnetic lens** focuses beam to Ø 10–200 µm; **deflection coils** scan/position beam
4. Power density ~ 10^{7}–10^{9} W/cm^{2} ⇒ vaporises in µs; pulses 4 µs–10 ms
5. Vacuum (≈ 10^{-5} torr) prevents scattering of electrons & oxidation of work

Process parameters

- Accelerating voltage 50–150 kV · Beam current 100 µA – 1 A (avg ~ 1–100 mA)
- Pulse duration 50 µs–10 ms, pulse frequency up to 16 kHz · Spot size 10–200 µm
- Lens current/focus, vacuum level, work material, feed (welding)

Power = V × I (e.g. 150 kV × 10 mA = 1.5 kW)

Applications

- **Micro-drilling** Ø 0.01–0.2 mm (spinnerets, filters, fuel nozzles, turbine cooling holes), slots, cutting thin foils/semiconductors, **electron-beam welding** (deep narrow weld, refractory metals), lithography (IC), annealing

Advantages	Limitations
Extremely fine holes, high precision; any material incl. diamond, W, ceramics; no mechanical force; fast (≈ 10^{4} holes/s); small HAZ; vacuum → no contamination	Very high capital cost; vacuum chamber (size limit, time to pump down); X-ray emission – shielding; skilled operator; thin sections only; recast layer

8. LBM vs PAM vs EBM

Item	LBM	PAM	EBM
Heat source	Photons (laser)	Ionised gas jet	Electrons
Medium	Air / assist gas	Ar/H_2/N_2 atmos.	Vacuum
Power density	~10^6–10^9 W/cm^2	~10^4–10^5 W/cm^2 (lowest)	Highest ~10^7–10^9 W/cm^2
Temp	~ 10,000 K local	10,000–30,000 K	~ 6000+ K local
Spot/kerf	0.01–0.5 mm	2–5 mm (large)	0.01–0.2 mm
Thickness	Thin–medium (≤ 25 mm)	Thick (≤ 150 mm)	Very thin (≤ 10 mm)
Accuracy	Good	Poor	Excellent
Work	All (non-reflective)	Conductors only	All
Cost	High	Low–medium	Very high
Main use	Drill, cut, mark	Plate cutting	Micro-holes, welding

9. NTM Process Selection Table

Process	Energy	Medium	Typical MRR	Best for
USM	Mechanical	Abrasive slurry	Low	Brittle/glass/ceramic
AJM / WJM	Mech. jet	Gas/water	Low	Cut, deburr, soft
CHM	Chemical	Etchant	Low	Shallow, sheet, large area
ECM	Electrochemical	Electrolyte	Very high	Hard conductive, cavities
EDM	Thermal (spark)	Dielectric	Medium	Dies, hard metals
LBM	Thermal	Air	Low–med	Micro-holes, cut
PAM	Thermal	Gas	Very high	Thick plate cut
EBM	Thermal	Vacuum	Low	Finest holes, welds

10. Quick Revision & Likely Questions

★ *Important*

- CHM steps: **Clean → Mask → Scribe → Etch → Demask**
- Maskants: neoprene, PVC, polyethylene, photoresist; etch factor = undercut / depth
- ECM: $MRR = \eta AI/(\rho FZ)$; gap 0.1–0.5 mm; V 5–25 V; NaCl/$NaNO_3$; Cu/brass tool, no wear
- ECG: 90% electrolysis + 10% abrasion · ECH: stones remove passive film
- Laser: stimulated emission + population inversion; Nd:YAG 1.064 μm (drill), CO_2 10.6 μm (cut)
- PAM: 10,000–30,000 K, transferred arc cuts SS/Al · EBM: 50–150 kV, vacuum, micro-holes

Likely exam questions

1. Explain chemical machining; steps, maskants, etchants, applications. (10M)
2. Define etch factor & undercut. How are maskants applied? (5M)
3. With neat sketch explain ECM; derive MRR; list advantages, limitations. (10M)
4. Discuss tool design & equilibrium gap in ECM. (5M)
5. Explain ECG and ECH; compare with conventional grinding/honing. (10M)
6. Describe laser principle & Nd:YAG / CO_2 lasers; laser drilling. (10M)
7. Explain plasma arc machining with sketch; transferred vs non-transferred. (10M)
8. Working of EBM with sketch; process parameters; applications. (10M)
9. Compare LBM, PAM, EBM; CHM vs ECM. (5M)
10. Numerical: MRR of ECM for given material, I, η.